Extremele funcțiilor cu două variabile

1. Fie $f: D = \mathbb{R}^2 \to \mathbb{R}$; $f(x,y) = x^2+xy+y^2-4\ln x - 10\ln y + 3$. Determinați punctele de extrem și calculați valorile funcției în aceste puncte.

Soluție

C.E. $\ln x$ și $\ln y$ trebuie să existe $\Rightarrow x>0$; $y>0 \Rightarrow$

$\Rightarrow D = \{(x,y) \in \mathbb{R}^2 \mid x>0; y>0\} = (0,\infty)\times(0,\infty)$

Punctele critice sunt soluțiile sistemului:

$$\begin{cases} \frac{\partial f}{\partial x} = 0 \\ \frac{\partial f}{\partial y} = 0 \end{cases} \Leftrightarrow \begin{cases} 2x+y-\frac{4}{x} = 0 \\ x+2y-\frac{10}{y} = 0 \Rightarrow x = \frac{10}{y} - 2y \end{cases} \Rightarrow$$

$$\Rightarrow 2\cdot\left(\frac{10}{y}-2y\right)+y-\frac{4}{\frac{10}{y}-2y} = 0$$

$$\frac{20}{y} - 4y + y - \frac{4y}{10-2y^2} = 0$$

$$\frac{20}{y} - 3y - \frac{2y}{5-y^2} = 0 \quad (y(5-y^2))$$

$$100 - 20y^2 - 15y^2 + 3y^4 - 2y^2 = 0$$

$$3y^4 - 37y^2 + 100 = 0$$

Notăm: $t = y^2 > 0$

$$\Rightarrow 3t^2 - 37t + 100 = 0$$

$$\Delta_t = 37^2 - 4\cdot 100\cdot 3 = 1369 - 1200 = 169 \Rightarrow \sqrt{\Delta_t} = 13$$

$$\Rightarrow t_{1,2} = \frac{37 \pm 13}{6} \rightarrow t_1 = \frac{50}{6} = \frac{25}{3}$$

$$t_2 = \frac{24}{6} = 4$$

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$pt: t = \frac{25}{3} \Rightarrow y = \sqrt{\frac{25}{3}} = \frac{5}{\sqrt{3}} = \frac{5\sqrt{3}}{3} \in D$

$pt: t = 4 \Rightarrow y = \sqrt{4} = 2 \in D.$

$pt\ y = \frac{5\sqrt{3}}{3} \Rightarrow x = \frac{10 \cdot \sqrt{3}}{5\sqrt{3}} - \frac{10\sqrt{3}}{3} = 2\sqrt{3} - \frac{10\sqrt{3}}{3} =$

$= \frac{-4\sqrt{3}}{3} < 0 \Rightarrow x_1 \notin D.$

$pt\ y = 2 \Rightarrow x = \frac{10}{2} - 4 = 5 - 4 = 1 \in D.$

$\Rightarrow$ singurul pct. critic este $(1,2)$.

$$H_{f(x,y)} = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{pmatrix} =$$

$\Delta_1 = \frac{\partial^2 f}{\partial x^2} = 2 + \frac{4}{x^2} = 2 + \frac{4}{1} = 6 > 0 \quad (1)$

$\Delta_2 = \det H$

$\frac{\partial^2 f}{\partial x \partial y} = 1 \; ; \frac{\partial^2 f}{\partial y \partial x} = 1 \; ; \frac{\partial^2 f}{\partial y^2} = 2 + \frac{10}{y^2} = 2 + \frac{10}{4} = \frac{9}{2}$

$\Delta_2 = \begin{vmatrix} 6 & 1 \\ 1 & 9/2 \end{vmatrix} = 6 \cdot \frac{9}{2} - 1 = 26 > 0 \quad (2)$

$\xRightarrow{(1)(2)}$ pct $(1,2)$ este pct de minim local

$f(1,2) = 1 + 2 + 4 - 4 \cdot \ln 1 - 10 \ln 2 + 3 =$

$= 7 - 0 - 10 \ln 2 + 3 =$

$= 10(1 - \ln 2)$

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2. Fie $f: D = \mathbb{R}^2 \to \mathbb{R}$; $f(x,y) = x^3+y^3-6xy$.

a) Pentru $D = \mathbb{R}^2$, determinați punctele de extrem și calculați valorile funcției în aceste puncte.

b) Pentru $D = \{(x,y) \in \mathbb{R}^2 \mid x \geq 0 ; y \geq 0 ; x+y \leq 5\}$ determinați valoarea minimă și maximă a funcției.

Soluție

a) Punctele critice sunt soluțiile sistemului:

$$\begin{cases} \frac{\partial f}{\partial x} = 0 \\ \frac{\partial f}{\partial y} = 0 \end{cases} \Leftrightarrow \begin{cases} 3x^2-6y=0 \\ 3y^2-6x=0 \end{cases} \Leftrightarrow \begin{cases} x^2-2y=0 \\ y^2-2x=0 \Rightarrow x = \frac{y^2}{2} \end{cases} \Rightarrow$$

$$\Rightarrow \left(\frac{y^2}{2}\right)^2 - 2y = 0 \Rightarrow \frac{y^4}{4} - 2y = 0 \Rightarrow y^4 - 8y = 0 \Rightarrow$$

$$\Rightarrow y(y^3-8) = 0 \Rightarrow y_1 = 0 ; y_2 = 2.$$

$$x_1 = \frac{y_1^2}{2} = \frac{0}{2} = 0 ; x_2 = \frac{y_2^2}{2} = \frac{4}{2} = 2$$

$\Rightarrow$ Obținem pct critice $(0,0)$ și $(2,2)$

$$H_{f(x,y)} = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{pmatrix}$$

$$\frac{\partial^2 f}{\partial x^2} = 6x ; \frac{\partial^2 f}{\partial x \partial y} = -6 ; \frac{\partial^2 f}{\partial y \partial x} = -6 ; \frac{\partial^2 f}{\partial y^2} = 6y$$

$$\Rightarrow H_{f(x,y)} = \begin{pmatrix} 6x & -6 \\ -6 & 6y \end{pmatrix}$$

caz I: pt pct $(0,0) \Rightarrow H_{f(0,0)} = \begin{pmatrix} 0 & -6 \\ -6 & 0 \end{pmatrix}$

$\Delta_1 = 0 \Rightarrow$ calculăm valorile proprii ale matricei $H_{f(0,0)}$.

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$\det\left(H_{f(0,0)} - \lambda I_2\right) = 0 \Leftrightarrow \begin{vmatrix} -\lambda - 6 & \\ -6 & -\lambda \end{vmatrix} = 0 \Leftrightarrow$

$\Leftrightarrow \lambda^2 - 36 = 0 \Leftrightarrow \lambda^2 = 36 \Rightarrow \lambda_1 = 6;\ \lambda_2 = -6$

Solutiile sunt de semne contrare $\Rightarrow$ pct $(0,0)$ nu este pct de extrem.

Cazul II: pt pct $(2,2) \Rightarrow H_{f(2,2)} = \begin{pmatrix} 12 & -6 \\ -6 & 12 \end{pmatrix}$

$\Delta_1 = 12 > 0$ (1)

$\Delta_2 = \det H_{f(2,2)} = \begin{vmatrix} 12 & -6 \\ -6 & 12 \end{vmatrix} = 36 \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} = 36(4-1) =$

$= 108 > 0$ (2)

(1)(2) $\Rightarrow$ pct $(2,2)$ este pct de minim local

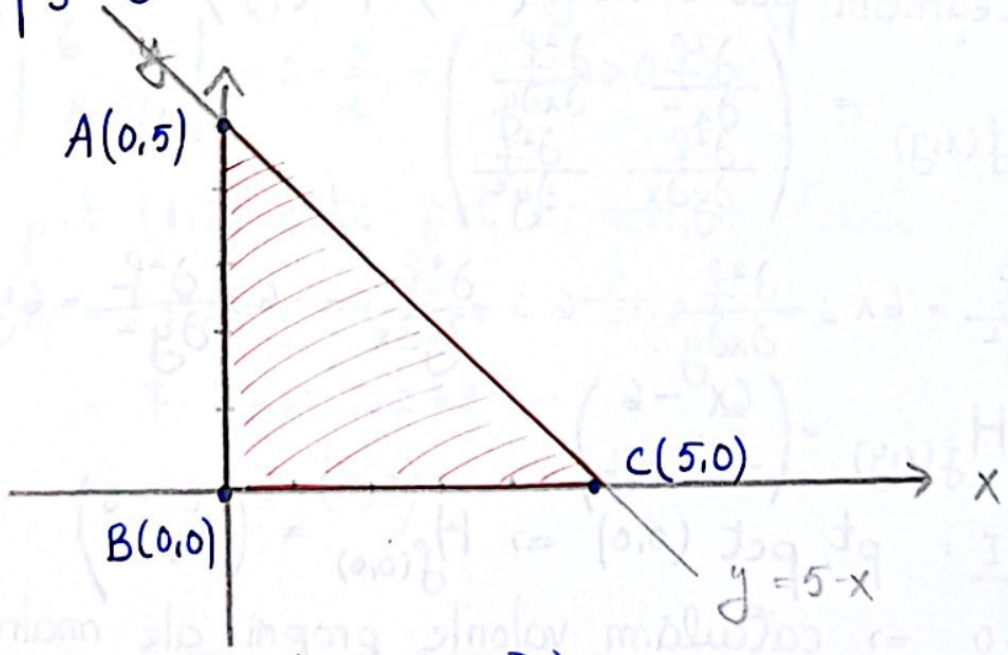
$f(2,2) = 8 + 8 - 24 = -8$.

b) Reprezentăm grafic domeniul D.

Avem $x + y \leq 5 \Rightarrow y \leq 5 - x$.

x	0	5
y	5	0

; $\text{Int } D = \{(x,y) \in \mathbb{R}^2 \mid x > 0; y > 0; x + y < 5\}$



— → Frontiera (Fr D)

/// → Interiorul (Int D)

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AB: $x = 0$; $y \in [0,5]$

$f(0,y) = y^3 \overset{not}{=} g_1(y)$

$g_1'(y) = 3y^2 = 0 \Leftrightarrow y = 0$

y	0		5
$g_1'(y)$	0	$+ + + + + + + + + + + + + + +$	
$g_1(y)$	0	$\nearrow$	125

$g_1(0) = f(0,0) = 0$

$g_1(5) = f(0,5) = 125$

BC: $y = 0$; $x \in [0,5]$

$f(x,0) = x^3 \overset{not}{=} g_2(x)$

$g_2'(x) = 3x^2 = 0 \Leftrightarrow x = 0$

x	0		5
$g_2'(x)$	0	$+ + + + + + + + + + + + + + +$	
$g_2(x)$	0	$\nearrow$	125

$g_2(0) = f(0,0) = 0$

$g_2(5) = f(5,0) = 125$

CA: $x \in [0,5]$; $y = 5 - x$

$f(x,y) = x^3 + (5-x)^3 - 6x(5-x) = \cancel{x^3} - \cancel{x^3} + 15x^2 - 75x + 125 - 30x + 6x^2 = 21x^2 - 105x + 125 \overset{not}{=} g_3(x)$

$g_3'(x) = 42x - 105 = 0 \Leftrightarrow x = \frac{105}{42} = \frac{5}{2}$

x	0		$\frac{5}{2}$		5
$g_3'(x)$	$-$	$- - - - - - - 0$		$+ + + + + +$	
$g_3(x)$	125	$\searrow$	$-\frac{25}{4}$	$\nearrow$	125

$g_3(0) = f(0,5) = 125$

$g_3\left(\frac{5}{2}\right) = f\left(\frac{5}{2}, \frac{5}{2}\right) = -\frac{25}{4}$

$g_3(5) = f(5,0) = 125$

Dim a) cunoaștem $(2,2)$ pct de min. local.

$(2,2) \in \text{Int } D \Rightarrow (2,2)$ pct de min; $f(2,2) = -8$

$\Rightarrow \max\limits_{x,y \in D} f(x,y) = 125$ și $\min\limits_{x,y \in D} f(x,y) = -8$.

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[3] Fie $f: D \subseteq \mathbb{R}^2 \to \mathbb{R}$: $f(x,y) = 3xy^2 - x^3 - 15x - 36y + 9$.

a) Pentru $D = \mathbb{R}^2$, determinați punctele de extrem și calculați valorile funcției în aceste puncte.

b) Pentru $D = [-4,4] \times [-3,3]$ determinați valoarea minimă și maximă a funcției.

Soluție

a) Pentru a găsi pct critice rezolvăm sist:

$$\begin{cases} \frac{\partial f}{\partial x} = 0 \\ \frac{\partial f}{\partial y} = 0 \end{cases} \Leftrightarrow \begin{cases} 3y^2 - 3x^2 - 15 = 0 \\ 6xy - 36 = 0 \Rightarrow x = \frac{6}{y} \end{cases} \Rightarrow$$

$$\Rightarrow 3y^2 - 3 \cdot \frac{36}{y^2} - 15 = 0 \mid \cdot y^2 \neq 0$$

$$3y^4 - 108 - 15y^2 = 0 \mid : 3$$

$$y^4 - 5y^2 - 36 = 0$$

Notăm: $y^2 = t > 0$

$$\Rightarrow t^2 - 5t - 36 = 0$$

$$\Delta_t = 25 + 144 = 169 \Rightarrow \sqrt{\Delta_t} = 13$$

$$t_{1,2} = \frac{5 \pm 13}{2} \rightarrow t_1 = \frac{18}{2} = 9$$

$$t_2 = \frac{-8}{2} = -4 < 0 \text{ (Fals)}$$

$$\Rightarrow t = 9 \Rightarrow y \in \{-3, 3\}$$

pt $y = -3 \Rightarrow x = -2 \Rightarrow (-2,-3)$ pct critic

pt $y = 3 \Rightarrow x = 2 \Rightarrow (2,3)$ pct critic.

$$H_{f(x,y)} = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{pmatrix}$$

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$\frac{\partial^2 f}{\partial x^2} = -6x$; $\frac{\partial^2 f}{\partial x \partial y} = -6y$; $\frac{\partial^2 f}{\partial y \partial x} = -6y$; $\frac{\partial^2 f}{\partial y^2} = 6x$

$\Rightarrow H_{f(x,y)} = \begin{pmatrix} -6x & 6y \\ 6y & 6x \end{pmatrix}$

caz I: pt $(-2,-3) \Rightarrow H_{f(-2,-3)} = \begin{pmatrix} 12 & -18 \\ -18 & -12 \end{pmatrix}$

$\Delta_1 = 12 > 0$ (1)

$\Delta_2 = \det H_{f(-2,-3)} = \begin{vmatrix} 12 & -18 \\ -18 & -12 \end{vmatrix} = 36 \begin{vmatrix} 2 & -3 \\ -3 & -2 \end{vmatrix} =$

$= 36(-4-9) < 0$ (2)

(1)(2) $\Rightarrow$ pct $(-2,-3)$ nu este pct de extrem.

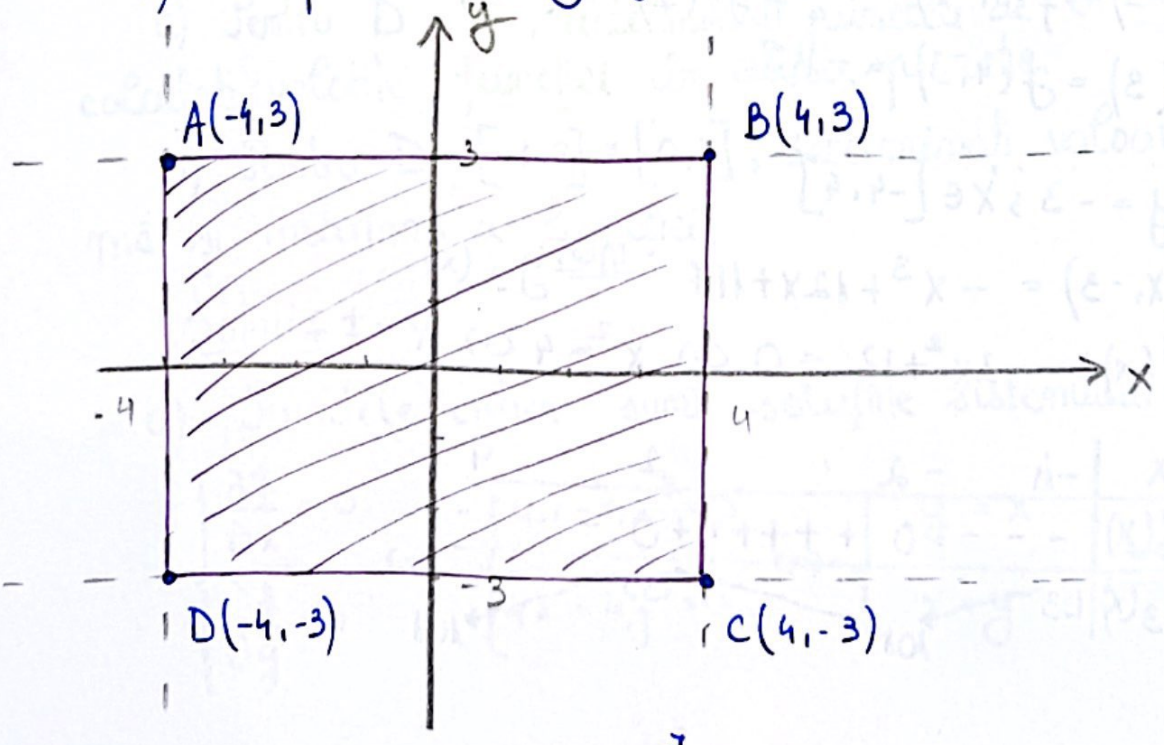
caz II: pt $(2,3) \Rightarrow H_{f(2,3)} = \begin{pmatrix} -12 & 18 \\ 18 & 12 \end{pmatrix}$

$\Delta_1 = -12 < 0$ (3)

$\Delta_2 = \det H_{f(2,3)} = \begin{vmatrix} -12 & 18 \\ 18 & 12 \end{vmatrix} = -144 - 324 < 0$ (4)

(3)(4) $\Rightarrow$ pct $(2,3)$ nu este pct de extrem.

b). Reprezentăm grafic domeniul D.



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AB: $y = 3$; $x \in [-4;4]$

$f(x;3) = -x^3 + 12x - 99 \xrightarrow{\text{not}} g_1(x)$

$g_1'(x) = -3x^2 + 12 = 0 \Leftrightarrow x^2 = 4 \Leftrightarrow x = \pm 2$

x	-4		-2		2		4
$g_1'(x)$	$-$	$-$	0	$+$	0	$-$	$-$
$g_1(x)$	-83	↘	-115	↗	-83	↘	-115

$g_1(-4) = f(-4;3) = -83$; $g_1(-2) = f(-2;3) = -115$;

$g_1(2) = f(2;3) = -83$; $g_1(4) = f(4;3) = -115$.

BC: $x = 4$; $y \in [-3;3]$

$f(4;y) = 12y^2 - 36y - 115 \xrightarrow{\text{not}} g_2(y)$

$g_2'(y) = 24y - 36 = 0 \Leftrightarrow y = \frac{3}{2}$

y	-3		$\frac{3}{2}$		3
$g_2'(y)$	$-$	$-$	0	$+$	$+$
$g_2(y)$	101	↘	-142	↗	-115

$g_2(-3) = f(4;-3) = 101$; $g_2\left(\frac{3}{2}\right) = f\left(4;\frac{3}{2}\right) = -142$;

$g_2(3) = f(4;3) = -115$.

CD: $y = -3$; $x \in [-4;4]$

$f(x;-3) = -x^3 + 12x + 117 \xrightarrow{\text{not}} g_3(x)$

$g_3'(x) = -3x^2 + 12 = 0 \Leftrightarrow x^2 = 4 \Leftrightarrow x = \pm 2$

x	-4		-2		2		4
$g_3'(x)$	$-$	$-$	0	$+$	0	$-$	$-$
$g_3(x)$	133	↘	101	↗	133	↘	101

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$g_3(-4) = f(-4,-3) = 133$; $g_3(-2) = f(-2,-3) = 101$;
$g_3(2) = f(2,-3) = 133$; $g_3(4) = f(4,-3) = 101$

DA: $x = -4$; $y \in [-3,3]$

$f(-4,y) = -12y^2 - 36y + 133 \overset{not}{=} g_4(x)$

$g_4'(x) = -24y - 36 = 0 \Leftrightarrow y = -\frac{3}{2}$

y	-3		$-\frac{3}{2}$		3
$g_4'(y)$	$++++$	0	$- - -$		
$g_4(y)$	133	↗	160	↘	-83

$g_4(-3) = f(-4,-3) = 133$; $g_4\left(-\frac{3}{2}\right) = f\left(-4;-\frac{3}{2}\right) = 160$;

$g_4(3) = f(-4,3) = -83$.

$\Rightarrow \max\limits_{x,y \in D} f(x,y) = 160$ și $\min\limits_{x,y \in D} f(x,y) = -142$.

4. Fie $f : D \subseteq \mathbb{R}^2 \to \mathbb{R}$; $f(x,y) = 4xy - x^4 - y^4$

a) Pentru $D = \mathbb{R}^2$, determinați punctele de extrem și calculați valorile funcției în aceste puncte.

b) Pentru $D = [-1,2] \times [0,2]$, determinați valoarea minimă și maximă a funcției.

Soluție

a) Punctele critice sunt soluțiile sistemului:

$$\begin{cases} \frac{\partial f}{\partial x} = 0 \\ \frac{\partial f}{\partial y} = 0 \end{cases} \Leftrightarrow \begin{cases} 4y - 4x^3 = 0 \\ 4x - 4y^3 = 0 \end{cases} \Leftrightarrow \begin{cases} y = x^3 \\ x = y^3 \end{cases} \Rightarrow$$

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$\Rightarrow x = (x^3)^3 = x^9 \Rightarrow x^9 - x = 0 \Rightarrow x(x^8-1) = 0 \Rightarrow x \in \{-1, 0, 1\}$

pt $x = -1 \Rightarrow y = -1 \Rightarrow (-1,-1)$ pct critic

pt $x = 0 \Rightarrow y = 0 \Rightarrow (0,0)$ pct critic

pt $x = 1 \Rightarrow y = 1 \Rightarrow (1,1)$ pct critic.

$$H_{f(x,y)} = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{pmatrix}$$

$\frac{\partial^2 f}{\partial x^2} = -12x^2$; $\frac{\partial^2 f}{\partial x \partial y} = 4$; $\frac{\partial^2 f}{\partial y \partial x} = 4$; $\frac{\partial^2 f}{\partial y^2} = -12y^2$

$$\Rightarrow H_{f(x,y)} = \begin{pmatrix} -12x^2 & 4 \\ 4 & -12y^2 \end{pmatrix}$$

<u>caz I</u>: pt $(-1,-1)$: $H_{f(-1,-1)} = \begin{pmatrix} -12 & 4 \\ 4 & -12 \end{pmatrix}$

$\Delta_1 = -12 < 0$ (1)

$\Delta_2 = \det H_{f(-1,-1)} = \begin{vmatrix} -12 & 4 \\ 4 & -12 \end{vmatrix} = 144 - 16 = 128 > 0$ (2)

<u>(1)(2)</u> $\Rightarrow$ pct $(-1,-1)$ este pct de maxim local

<u>caz II</u> : pt $(0,0)$: $H_{f(0,0)} = \begin{pmatrix} 0 & 4 \\ 4 & 0 \end{pmatrix}$

$\Delta_1 = 0 \Rightarrow$ calculăm valorile proprii ale matricei $H_{f(0,0)}$.

$$\det(H_{f(0,0)} - \lambda I_2) = \begin{vmatrix} -\lambda & 4 \\ 4 & -\lambda \end{vmatrix} = 0 \Leftrightarrow$$

$\Leftrightarrow \lambda^2 - 16 = 0 \Leftrightarrow \lambda^2 = 16 \Leftrightarrow \lambda_1 = 4$ și $\lambda_2 = -4$

Soluțiile au semne contrare $\Rightarrow (0,0)$ nu este pct de extrem.

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caz III: pct $(1,1)$: $H_{f(1,1)} = \begin{pmatrix} -12 & 4 \\ 4 & -12 \end{pmatrix}$

$\Delta_1 = -12 < 0$ (3)

$\Delta_2 = \det H_{f(1,1)} = \begin{vmatrix} -12 & 4 \\ 4 & -12 \end{vmatrix} > 0$ (4)

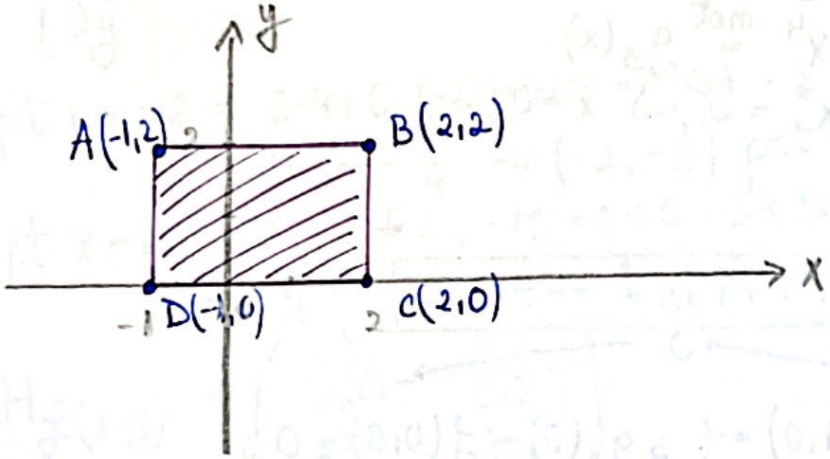
(3)(4) $\Rightarrow$ pct $(1,1)$ este pct de maxim local

$\Rightarrow$ $f(-1,-1) = 4 - 1 - 1 = 2$

$f(1,1) = 4 - 1 - 1 = 2$

b) Reprezentăm grafic domeniul D

Pentru $\text{Int } D = (-1,2) \times (0,2)$, avem pct $(1,1)$ ca punct de maxim, cu $f(1,1) = 2$.



AB: $y = 2$; $x \in [-1,2]$

$f(x,2) = -x^4 + 8x - 16 \xrightarrow{not} g_1(x)$

$g_1'(x) = -4x^3 + 8 = 0 \Leftrightarrow x^3 = 2 \Leftrightarrow x = \sqrt[3]{2}$

x	-1		$\sqrt[3]{2}$		2
$g_1'(x)$	$+$	$+$ $+$ $+$ $+$ $+$ $+$ $+$ $+$	0	$-$ $-$ $-$ $-$	
$g_1(x)$	-25	$\nearrow$	$6\sqrt[3]{2} - 16$	$\searrow$	-16

$g_1(-1) = f(-1,2) = -25$; $g_1(\sqrt[3]{2}) = f(\sqrt[3]{2},2) = 6\sqrt[3]{2} - 16$

$g_1(2) = f(2,2) = -16$

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BC: $x = 2$; $y \in [0,2]$

$f(2,y) = -y^4 + 8y - 16 \overset{\text{not}}{=} g_2(y)$

$g_2'(y) = -4y^3 + 8 = 0 \Leftrightarrow y = \sqrt[3]{2}$

y	0		$\sqrt[3]{2}$		2
$g_2'(y)$	$+$	$+$	0	$-$	$-$
$g_2(y)$	-16	↗	$6\sqrt[3]{2} - 16$	↘	-16

$g_2(0) = f(2,0) = -16$; $g_2(\sqrt[3]{2}) = f(2, \sqrt[3]{2}) = 6\sqrt[3]{2} - 16$

$g_2(2) = f(2,2) = -16$

CD: $y = 0$; $x \in [-1,2]$

$f(x,0) = -x^4 \overset{\text{not}}{=} g_3(x)$

$g_3'(x) = -4x^3 = 0 \Leftrightarrow x = 0$

x	-1		0		2
$g_3'(x)$	$+$	$+$	0	$-$	$-$
$g_3(x)$	-1	↗	0	↘	-16

$g_3(-1) = f(-1,0) = -1$; $g_3(0) = f(0,0) = 0$;

$g_3(2) = f(2,0) = -16$

DA: $x = -1$; $y \in [0,2]$

$f(-1,y) = -y^4 - 4y - 1 \overset{\text{not}}{=} g_4(y)$

$g_4'(y) = -4y^3 - 4 = 0 \Leftrightarrow y^3 = -1 \Leftrightarrow y = -1 \notin [0,2]$

y	0		2
$g_4(y)$	-1	↘	-25

$g_4(0) = f(-1,0) = -1$; $g_4(2) = f(-1,2) = -25$

$\Rightarrow \max\limits_{x,y \in D} f(x,y) = 2$; $\min\limits_{x,y \in D} f(x,y) = -25$

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5. Fie $f: D \subseteq \mathbb{R}^2 \to \mathbb{R}$; $f(x,y) = x^3 + 3x^2y - 15x - 12y$

a) Pentru $D = \mathbb{R}^2$, determinați punctele de extrem și calculați valorile funcției în aceste puncte.

b) Pentru $D = \{(x,y) \in \mathbb{R}^2 \mid x \geq 0; y \geq 0; 3y + x \leq 3\}$ determinați valoarea minimă și maximă a funcției.

Soluție:

a) Pentru a identifica punctele critice rezolvăm sistemul:

$$\begin{cases} \frac{\partial f}{\partial x} = 0 \\ \frac{\partial f}{\partial y} = 0 \end{cases} \Leftrightarrow \begin{cases} 3x^2 + 6xy - 15 = 0 \\ 3x^2 - 12 = 0 \Leftrightarrow x^2 = 4 \Leftrightarrow x \in \{-2, 2\} \end{cases}$$

pt $x = -2 \Rightarrow 3 \cdot 4 + 6 \cdot (-2)y - 15 = 0 \Leftrightarrow -3 - 12y = 0 \Leftrightarrow$

$\Leftrightarrow y = -\frac{1}{4} \Rightarrow (-2, -\frac{1}{4})$ pct critic

pt $x = 2 \Rightarrow 3 \cdot 4 + 6 \cdot 2y - 15 = 0 \Leftrightarrow -3 + 12y = 0 \Leftrightarrow$

$\Leftrightarrow y = \frac{1}{4} \Rightarrow (2, \frac{1}{4})$ pct critic.

$$H_{f(x,y)} = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{pmatrix}$$

$\frac{\partial^2 f}{\partial x^2} = 6x + 6y$; $\frac{\partial^2 f}{\partial x \partial y} = 6x$; $\frac{\partial^2 f}{\partial y \partial x} = 6x$; $\frac{\partial^2 f}{\partial y^2} = 0$

$$\Rightarrow H_{f(x,y)} = \begin{pmatrix} 6x+6y & 6x \\ 6x & 0 \end{pmatrix}$$

caz I: pt $(-2, -\frac{1}{4})$: $H_{f(-2,-\frac{1}{4})} = \begin{pmatrix} -27/2 & -12 \\ -12 & 0 \end{pmatrix}$

$\Delta_1 = -\frac{27}{2} < 0$ (1)

$\Delta_2 = \det H_{f(-2,-\frac{1}{4})} = \begin{vmatrix} -27/2 & -12 \\ -12 & 0 \end{vmatrix} = -144 < 0$ (2)

(1)(2) $\Rightarrow$ pct $(-2, -\frac{1}{4})$ nu este pct de extrem

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caz II: pt $(2, \frac{1}{4})$: $H_{f(2,\frac{1}{4})} = \begin{pmatrix} 27/2 & 12 \\ 12 & 0 \end{pmatrix}$

$\Delta_1 = \frac{27}{2} > 0$ (3)

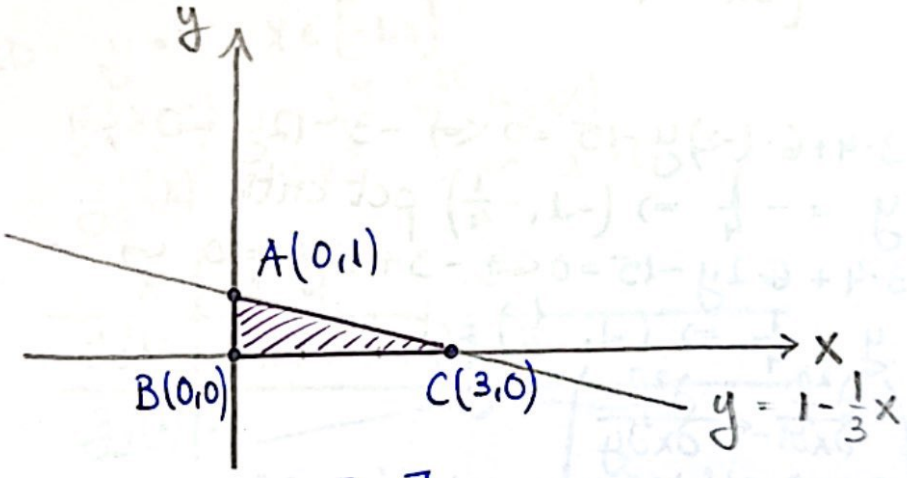
$\Delta_2 = \det H_{f(2,1/4)} = \begin{vmatrix} 27/2 & 12 \\ 12 & 0 \end{vmatrix} = -144 < 0$ (4)

(3)(4) $\Rightarrow$ pct $(2, \frac{1}{4})$ nu este pct de extrem.

b) Reprezentăm grafic domeniul D.

Avem: $3y + x \leq 3 \Leftrightarrow y \leq 1 - \frac{1}{3}x$

x	0	3
y	1	0



AB: $x = 0$; $y \in [0,1]$

$f(0,y) = -12y \xrightarrow{not} g_1(y)$

y	0		1
$g_1(y)$	0	$\searrow$	-12

$g_1(0) = f(0,0) = 0$; $g_1(1) = f(0,1) = -12$.

BC: $y = 0$; $x \in [0,3]$

$f(x,0) = x^3 - 15x \xrightarrow{not} g_2(x)$

$g_2'(x) = 3x^2 - 15 = 0 \Leftrightarrow x^2 = 5 \Leftrightarrow x = \pm\sqrt{5}$ $(-\sqrt{5} \notin [0,3])$

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x	0		$\sqrt{5}$		3
$g_2'(x)$	$-$	$-$	0	$+$	$+$
$g_2(x)$	0	$\searrow$	$-10\sqrt{5}$	$\nearrow$	-18

$g_2(0) = f(0,0) = 0$; $g_2(\sqrt{5}) = f(\sqrt{5},0) = -10\sqrt{5}$;

$g_2(3) = f(3,0) = -18$

CA: $x \in [0,3]$; $y = 1 - \frac{1}{3}x$

$f(x,y) = f(x, 1-\frac{1}{3}x) = x^3 + 3x^2\left(1-\frac{1}{3}x\right) - 15x - 12\left(1-\frac{1}{3}x\right) =$

$= x^3 + 3x^2 - x^3 - 15x - 12 + 4x = 3x^2 - 11x - 12 \overset{not}{=} g_3(x)$

$g_3'(x) = 6x - 11 = 0 \Leftrightarrow x = \frac{11}{6}$

x	0		$\frac{11}{6}$		3
$g_3'(x)$	$-$	$-$	0	$+$	$+$
$g_3(x)$	-12	$\searrow$	$-\frac{265}{12}$	$\nearrow$	-18

$g_3(0) = f(0,1) = -12$; $g_3\left(\frac{11}{6}\right) = f\left(\frac{11}{6}, \frac{7}{18}\right) = -\frac{265}{12}$

$g_3(3) = f(3,0) = -18$

$\Rightarrow \max\limits_{x,y \in D} f(x,y) = 0$; $\min\limits_{x,y \in D} f(x,y) = -10\sqrt{5}$.

6. Fie $f: [0,1] \times [0,1] \to \mathbb{R}$; $f(x,y) = xy(1-x-y)$. Să se determine valoarea minimă și maximă a funcției pe domeniul dat.

Pentru a afla punctele critice, rezolvăm sistemul:

$$\begin{cases} \frac{\partial f}{\partial x} = 0 \\ \frac{\partial f}{\partial y} = 0 \end{cases} \Leftrightarrow \begin{cases} y - 2xy - y^2 = 0 \mid : y \\ x - x^2 - 2xy = 0 \end{cases} \Rightarrow 1 - 2x - y = 0 \to y = 1 - 2x$$

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$\Rightarrow x - x^2 - 2x(1-2x) = 0$

$x - x^2 - 2x + 4x^2 = 0$

$3x^2 - x = 0 \rightarrow x_1 = 0 \Rightarrow y_1 = 1$

$x_2 = \frac{1}{3} \Rightarrow y_2 = \frac{1}{3}$

Avem $\text{Int } D = \{(x,y) \in \mathbb{R}^2 \mid x \in (0,1); y \in (0,1)\} \Rightarrow$

$\Rightarrow$ pct $(0,1) \notin \text{Int } D$; pct $\left(\frac{1}{3}, \frac{1}{3}\right) \in \text{Int } D$ și este punct critic.

$$H_{f(x,y)} = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{pmatrix} = \begin{pmatrix} -2x & 1-2x-2y \\ 1-2x-2y & -2x \end{pmatrix}$$

$$\Rightarrow H_{f(1/3;1/3)} = \begin{pmatrix} -2/3 & -1/3 \\ -1/3 & -2/3 \end{pmatrix}$$

$\Delta_1 = -\frac{2}{3} < 0$ (1)

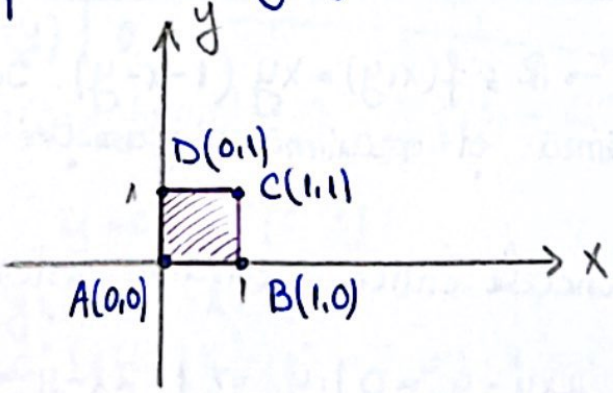
$$\Delta_2 = \det H_{f(1/3;1/3)} = \begin{vmatrix} -2/3 & -1/3 \\ -1/3 & -2/3 \end{vmatrix} = \frac{1}{9}\begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} = \frac{1}{9}(4-1) =$$

$= \frac{3}{9} = \frac{1}{3} > 0$ (2)

$\overset{(1)(2)}{\Longrightarrow}$ pct $\left(\frac{1}{3}, \frac{1}{3}\right)$ este pct de maxim local

$f\left(\frac{1}{3}, \frac{1}{3}\right) = \frac{1}{9} - \frac{1}{27} - \frac{1}{27} = \frac{1}{27}$

Reprezentăm grafic domeniul D.



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AB: $y = 0 ; x \in [0,1]$

$f(x,0) = 0$

BC: $x = 1 ; y \in [0,1]$

$f(1,y) = y - y - y^2 = -y^2 \overset{not}{=} g_1(y)$

$g_1'(y) = -2y = 0 \Leftrightarrow y = 0$

y	0		1
$g_1'(y)$	0	− − − − − − − − − −	
$g_1(y)$	0	⟶	-1

$g_1(0) = 0 = f(1,0) ; g_1(1) = f(1,1) = -1$

CD: $y = 1 ; x \in [0,1]$

$f(x,1) = x - x^2 - x = -x^2 = g_2(x)$

$g_2'(x) = -2x = 0 \Leftrightarrow x = 0$

x	0		1
$g_2'(x)$	0	− − − − − − − − −	
$g_2(x)$	0	⟶	-1

$g_2(0) = f(0,1) = 0 ; g_2(1) = f(1,1) = -1$

DA: $x = 0 ; y \in [0,1]$

$f(0,y) = 0$

$\Rightarrow \max\limits_{x,y \in D} f(x,y) = \frac{1}{27} ; \min\limits_{x,y \in D} f(x,y) = -1$

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